

A Resident-Streaming Roofline Model for Compute-in-Memory

Hanyong Shao^{1,2}, Kechao Tang^{†1,3}, and Ru Huang^{†1,3}

¹School of Integrated Circuits & ²Academy for Advanced Interdisciplinary Studies, Peking University, Beijing 100871, China ³Beijing Advanced Innovation Center for Integrated Circuits, Beijing 100871, China [†]Corresponding authors Email: {tkch, ruhuang}@pku.edu.cn

Abstract—From CPUs to GPUs, the Roofline model has become a textbook tool for revealing a platform’s performance ceiling and how well a workload matches it. Compute-in-memory (CIM), however, falls outside this picture because it violates the core Roofline assumption of separate compute and memory. We present a resident-streaming Roofline model for CIM, in which streaming-activation and resident-weight throughputs replace compute throughput and memory bandwidth, respectively. This follows from a symmetry we observe: conventional processors pair heterogeneous hardware with homogeneous data, whereas CIM pairs homogeneous hardware with heterogeneous data—operands residing in the array and operands streaming through it. Using NeuroSim with literature-extracted parameters, we place ten CIM technologies, from SRAM to emerging non-volatile memories, on this Roofline and quantify the demands of mainstream large language models across computation stages. The model explains why write speed is critical in CIM: like GPU bandwidth, it caps throughput whenever reuse falls below a critical value, which spans over four orders of magnitude across these memory technologies. Likewise, the throughput at which activations stream through the CIM macro plays the role of GPU compute throughput and represents the true computing performance of CIM.

Index Terms—Compute-in-memory, roofline model, streaming intensity, data reuse, large language models.

I. INTRODUCTION

The textbook Roofline model [1] relates hardware capability to workload demand through the Arithmetic Intensity (AI), and exposes the bottleneck arising from a mismatch between software and hardware. It describes the capability limits on computation and on data supply by the compute capability π and the memory bandwidth β , respectively. AI represents the number of operations a workload supports per byte of data transferred; comparing it with the hardware ridge point $AI^* = \pi/\beta$ distinguishes the memory-bound region from the compute-bound region and provides theoretical guidance for optimization [2], [3].

The validity of this model rests on an assumption implicit in the von Neumann architecture: memory and compute are separate units [4]. Fig. 1(a) shows the familiar picture. Storage and computation are carried out by different hardware, and although activations A and weights W play different roles in the algorithm, the hardware treats both simply as operands delivered to the compute units: neither has a distinct identity, and the two are interchangeable.

However, this assumption no longer holds in Compute-in-Memory (CIM), a non-von Neumann architecture [5]. In Fig. 1(b), the CIM macro that stores W also calculates A , so the two hardware units are unified [6]. The *weights* take part in computation as *resident state*, while each new *activation* enters through the read path and *streams across* the macro, where it is calculated against that state. CIM turns one class of operands, the weights, into stored state at the very site of

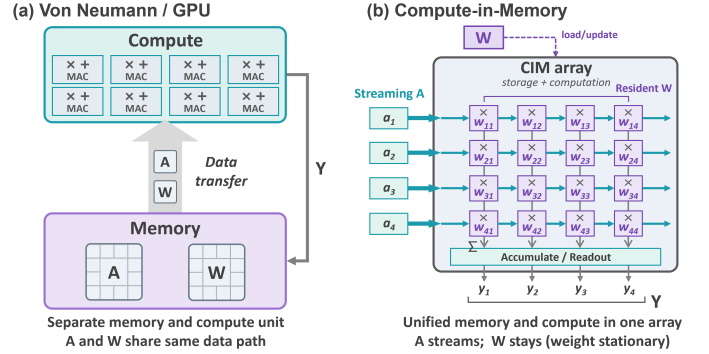


Fig. 1. Operand roles in (a) separate memory and compute units and (b) a CIM array. Streaming inputs A are evaluated against resident weights W , with a distinct path for loading or updating W .

calculation, while the other class, the activations, flows through the macro as inputs served by that state. The separate hardware units of memory and compute in CPU/GPU disappear, but new roles of streaming and resident operands emerge in CIM.

This change calls for a new Roofline model that rethinks how compute capability and data transfer relate in CIM. The need has two roots. First, the hardware fusion of the CIM architecture dissolves the concept of compute capability π : the capability of a CIM macro to complete multiply-accumulate (MAC) is, in essence, the “transfer rate” of A . Second, the split of operand roles creates two distinct throughputs: weights enter through the write path, which erases, programs, or refreshes the cells, whereas activations enter through the read path, which applies them as inputs and accumulates their products with the weights as outputs. Hence, whereas AI captures the GPU bottleneck as an imbalance between computation and memory access, the CIM bottleneck is the imbalance between these two operand throughputs. Prior studies have recognized this problem [7] and proposed several Roofline analyses for CIM [2], [8]–[14], yet none of them reveals this difference.

Therefore, we propose a resident-streaming Roofline model for CIM. Within an explicit workload window, the model separately counts the input bytes Q_S consumed by streaming calculation and the write bytes Q_R that update or reload resident state, and defines the Streaming Intensity $SI = Q_S/Q_R$ as the streaming demand supported per byte of resident write. The resulting demand–throughput pairing corresponds directly to the classical Roofline: the streaming throughput ρ plays the role of π , the resident throughput τ that of β , and SI, like AI, links the reuse behavior of a workload to the two hardware throughputs. By explicitly distinguishing the two roles of operands in computation, streaming A and resident W , the model turns the performance imbalance between resident updates and streaming calculation into a throughput bound and a critical reuse requirement.

Building on this model, we carry out a quantitative analysis from two sides: hardware capability and algorithm demand. On the hardware side, we characterize the streaming and resident throughputs of ten CIM technologies through NeuroSim simulations calibrated with parameters extracted from the literature. On the algorithm side, we examine the matrix multiplications in mainstream large language models (LLM) and characterize the demands posed by static model weights and by the dynamic key-value (KV) cache. This Roofline model provides clearer data-reuse requirements and hardware design guidance for heterogeneous computing across CIM technologies.

II. A RESIDENT-STREAMING ROOFLINE

A. Operand Roles and Counting Boundaries

Outline. Define an analysis window with logical streaming-input demand Q_S and resident-write demand Q_R , in bytes, at the boundaries sketched in Fig. 1. Count initial loading only when it occurs inside the window, include every update or reload, and match ρ and τ to the same configuration, precision, and service boundaries. Explain that resident state may be weights or historical keys and values; physical encoding, conversion, and required maintenance belong in service accounting.

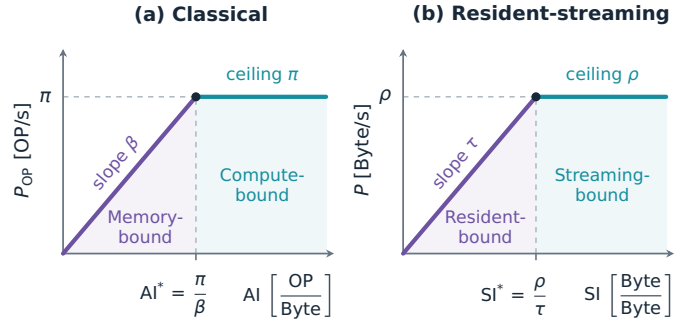
$$SI = \frac{Q_S}{Q_R}, \quad P = \frac{Q_S}{T} \quad [\text{Byte/s}], \quad (1)$$

where T is the window completion time; for $Q_S > 0$ and $Q_R = 0$, define $SI = \infty$.

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Counting boundaries: one compact definition paragraph and a window example.

Writing round 1: prose not yet drafted.



Aspect	GPU (a)	CIM (b)
Decomposition	Hardware resources	Data roles
Workload	OP, Byte	Q_S, Q_R
Service rates	π [OP/s], β [Byte/s]	ρ, τ [Byte/s]
Intensity	$AI = \text{OP/Byte}$	$SI = Q_S/Q_R$
Vertical axis	$P_{OP} = \text{OP}/T$	$P = Q_S/T$
Upper bound	$P_{OP} \leq \min(\pi, \beta AI)$	$P \leq \min(\rho, \tau SI)$
Left regime	Memory-bound	Resident-bound
Right regime	Compute-bound	Streaming-bound
Ridge	$AI^* = \pi/\beta$	$SI^* = \rho/\tau$

Fig. 2. Classical GPU [1] and resident-streaming CIM Rooflines with their accounting correspondence. Panels use independent linear scales; curves are upper bounds. OP and Byte denote operation and transferred-byte counts.

B. Throughput Bound and Streaming Intensity

Outline. Derive the two necessary service times and use Fig. 2 to distinguish the meaning of the axes from the familiar Roofline geometry. Identify resident-bound and streaming-bound regions by the tighter term, while treating the curve as an upper bound whose attainability depends on shared resources and scheduling. Use the two compact accounting examples below to connect input throughput to useful operations and expose the effect of changing the residency window.

$$T \geq \max\left(\frac{Q_S}{\rho}, \frac{Q_R}{\tau}\right), \quad (2)$$

$$P \leq \min(\rho, \tau SI), \quad SI^* = \frac{\rho}{\tau}. \quad (3)$$

Here ρ and τ are the streaming and resident throughputs in Byte/s; $SI < SI^*$ is resident-bound and $SI > SI^*$ is streaming-bound.

a) Example to develop: useful-operation conversion:

One logical 128×128 INT8 matrix-vector multiplication takes 128 input bytes and performs 16,384 MACs. With two useful operations per MAC, the same execution satisfies

$$\frac{32,768 \text{ OP}}{128 \text{ Byte}} = 256 \text{ OP/Byte}, \quad P_{OP} = 256P. \quad (4)$$

b) Example to develop: three residency windows: For the same matrix and U useful input vectors, $Q_S = 128U$ bytes; only the in-window loading policy changes:

Policy	Q_R [Byte]	SI
Already resident	0	∞
Load once, reuse	16,384	$U/128$
Reload before each use	$16,384U$	$1/128$

(5)

TABLE I
STREAMING INTENSITY AFTER ONE COMPLETE WEIGHT LOAD

Weight shape $N \times K$	Reuse count U					
	128	1K	16K	128K	1M	$\rightarrow \infty$
128×128	1.0	8.0	128.0	1024.0	8192.0	∞
1024×1024	0.125	1.0	16.0	128.0	1024.0	∞
4096×4096	0.0313	0.250	4.0	32.0	256.0	∞
1024×4096	0.125	1.0	16.0	128.0	1024.0	∞
4096×1024	0.0313	0.250	4.0	32.0	256.0	∞

One byte/element: $Q_S = UK$, $Q_R = NK$, $SI = U/N$. As $U \rightarrow \infty$, the single load $Q_R = NK$ stays finite. Shapes are logical matrices, not hardware tiles. 1K = 1024, 1M = 1024².

C. From Streaming Intensity to Reuse

Outline. Define U as all input vectors served after one complete load, including the first use and allowing accumulation across requests. Use Table I and the payload relations below to connect logical matrix shape and reuse, then translate the ridge into Fig. 4 at each native service boundary. Distinguish unbounded reuse of a finite initial load from a finite, already-resident window with no writes.

$$Q_S = UKb_S, \quad Q_R = NKb_R, \quad (6)$$

$$SI = \frac{Ub_S}{Nb_R}, \quad \rho = \frac{Kb_S}{\Delta_S}, \quad (7)$$

$$\tau = \frac{NKb_R}{T_R}, \quad U^* = \frac{T_R}{\Delta_S} = \frac{Nb_R}{b_S} SI^*. \quad (8)$$

The precisions b_S, b_R are bytes per element; Δ_S is a complete-vector service interval and T_R a full-matrix load service time. At equal byte widths, $SI = U/N$ and $U^* = NSI^*$.

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Reuse interpretation and the transition to native reference services.

Writing round 1: prose not yet drafted.

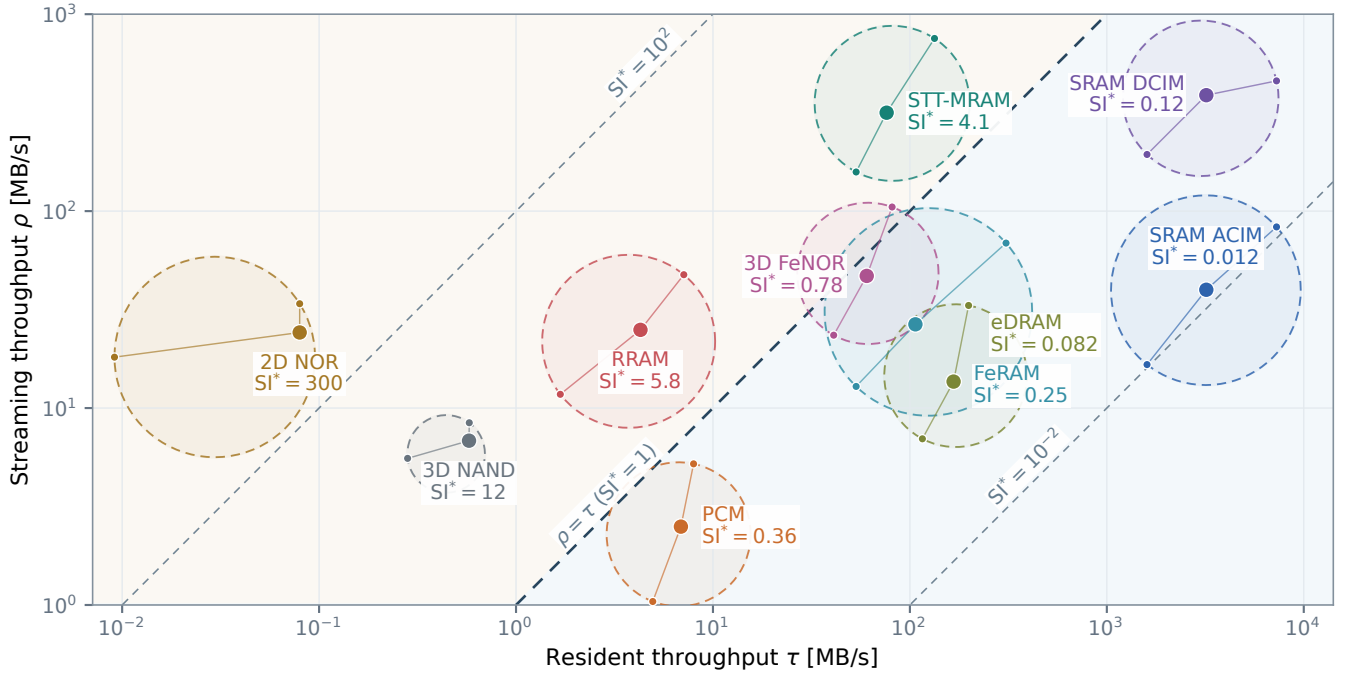


Fig. 3. Streaming throughput ρ and resident throughput τ for ten native reference configurations; $1 \text{ MB/s} = 10^6 \text{ Byte/s}$. Large points are typical scenarios; small points preserve the original paired alternatives. Circles summarize only those finite scenarios, and $SI^* = 1$ is a throughput-ratio reference, not a universal workload boundary. Native sizes and resources differ.

TABLE II
STREAMING INTENSITIES OF REPRESENTATIVE INFERENCE MATRIX STAGES

Model	Weight projections						Attention					
	QKV (U)			FFN/MoE (U)			Prefill (L)			Decode (L)		
	1K	128K	1M	16	1K	16K	1K	8K	64K	1K	8K	64K
Qwen3.5-2B	0.200	25.6	204.8	0.00347	0.222	3.6	6.0	34.0	258.0	10.0	66.0	514.0
Ministral 3 8B	0.167	21.3	170.7	0.00167	0.107	1.7	10.0	66.0	514.0	18.0	130.0	1026.0
Qwen3.6-35B-A3B	0.111	14.2	113.8	0.0130	0.833	13.3	12.0	68.0	516.0	20.0	132.0	1028.0
Hy3 (295B)	0.100	12.8	102.4	0.00477	0.306	4.9	20.0	132.0	1028.0	36.0	260.0	2052.0
Ling-1T (1000B)	0.100	12.8	102.4	0.00326	0.208	3.3	20.0	132.0	1028.0	36.0	260.0	2052.0
MiMo-V2.5-Pro (1020B)	0.0377	4.8	38.6	0.00347	0.222	3.6	35.2	214.4	1648.0	60.8	419.2	3286.4

Entries are SI; all roles use one byte/element. QKV includes any gate, excluding O. FFN/MoE counts one dense FFN or one routed expert. Prefill builds KV from empty; decode appends one token to a visible length L . 1K = 1024, 1M = 1024². Ling-1T at 64K uses the fixed extension.

III. HARDWARE AND WORKLOAD CHARACTERIZATION

A. Memory-Based Reference Configurations

Outline. Introduce the ten native analog/digital reference configurations in Fig. 3, simulated with NeuroSim [15], [16] from device and macro sources—SRAM DCIM [17]–[19], SRAM ACIM [7], [20], [21], 2D NOR [22]–[24], 3D NAND [25]–[27], RRAM [28]–[30], MRAM [31]–[33], PCM [34]–[36], FeRAM [37]–[39], eDRAM [40]–[42], and FeNOR [43]–[45]—and shared peripheral assumptions [46]–[48]. Explain the three paired engineering scenarios and complete service accounting, retaining native matrix dimensions, resources, and update organizations. Read absolute throughput and throughput ratio separately; these are literature-grounded estimates rather than equal-area chip comparisons.

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Native configurations and complete service accounting.

Writing round 1: prose not yet drafted.

B. Inference Matrix Workloads

Outline. Use the six fixed model configurations [49]–[54] as structural examples for Table II, counting common inputs once per matrix stage at one byte per element. Explain QKV intensity U/N_{proj} and FFN intensity $U(D + F)/(3DF)$ using output width N_{proj} (including any gate), hidden width D , and dense-FFN or single-expert width F . For grouped-query attention, use the native head sharing and possibly unequal query/key and value widths: prefill creates each token’s KV once, whereas decode appends one token and treats historical KV as resident.

IV. REUSE REQUIREMENTS AND DESIGN IMPLICATIONS

A. Critical Reuse Across Configurations

Outline. Use Fig. 4 to show that typical full-load thresholds span about 1.6 to 38,867 input vectors, then develop one concrete crossing: at the illustrative reuse $U = 128$, STT-MRAM is resident-bound in the short and typical scenarios (about 181 and 133), but streaming-bound in the long scenario (about 95). Connect the native threshold to Qwen3.5-2B and MiMo-V2.5-Pro QKV stages using the existing sequential full-tile service reference, for which both sides accumulate the same tile count. Keep this mapping as a component-service bound, with output reduction and other system costs outside its scope.

For native 128×128 tiles, let $m_K = D/128$, $m_N = N_{\text{proj}}/128$, and $M = m_K m_N$:

$$T_{S,\text{comp}} = MU\Delta_S, \quad T_{R,\text{comp}} = MT_R, \quad (9)$$

$$\rho_{\text{eff}} = \rho/m_N, \quad \tau_{\text{eff}} = \tau, \quad (10)$$

$$\text{SI}_{\text{eff}}^* = \frac{\text{SI}^*}{m_N} = \frac{U^*}{N_{\text{proj}}}. \quad (11)$$

The factor M cancels at service balance. This reference uses one engine, full tiles without padding, and equal U per tile; input replay is included in $T_{S,\text{comp}}$.

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Explain the threshold span and one scenario crossing.

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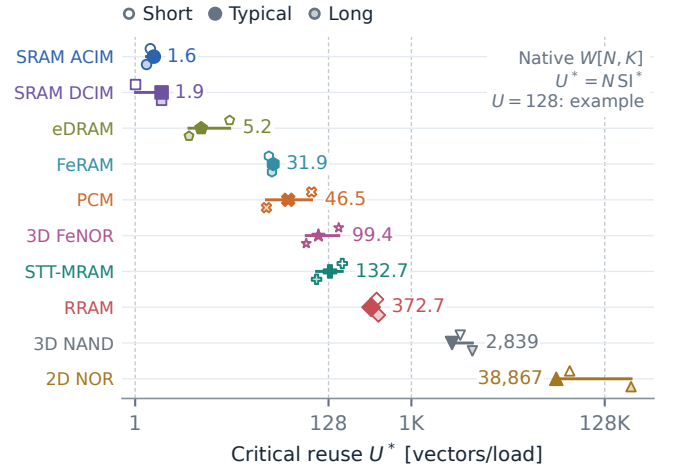


Fig. 4. Full-load critical reuse $U^* = T_R/\Delta_S = \text{NSI}^*$ at equal byte widths; each row uses its native $W[N, K]$. Numbers give typical values; segments span only three paired scenarios. Dashed lines mark illustrative reuse counts, not universal balance points. 1K = 1024.

B. Design Implications

Outline. Turn a target reuse count into an allowable full-load service time using (12), making Fig. 4 a design requirement rather than a ranking. Revisit Fig. 3 to explain why a lower threshold alone does not imply higher performance: streaming throughput still sets the upper plateau. Contrast weight residency with growing attention state in Table II; KV append requires its own update-service boundary before applying any full-load threshold.

$$T_R \leq U_{\text{target}} \Delta_S \iff U_{\text{target}} \geq U^*. \quad (12)$$

This is a necessary service-balance condition for reaching the streaming ceiling, not a guarantee of realized throughput.

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Turn a target reuse count into a resident-service requirement.

Writing round 1: prose not yet drafted.

V. CONCLUSION

Outline. Close with the role-based connection between in-window resident updates and streaming evaluation established by (3). Use Fig. 4 to emphasize the practical outcome: hardware service asymmetry becomes an explicit reuse requirement for residency and update decisions.

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Final conclusion: one short paragraph.

Writing round 1: prose not yet drafted.

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